

VITyarthi FLIPPED COURSE EVALUATION REPORT

BUILD YOUR OWN PROJECT: INTERACTIVE CLI NUMBER GUESSING GAME

Course: Problem Solving and Programming (Python)

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Degree: B.Tech Computer Science	Date of Submission: _____

Submitted in partial fulfillment of the academic requirements for Python Programming Evaluation.

2. Introduction

The **Interactive CLI Number Guessing Game** is a console-driven software project developed as part of the flipped course evaluation under VITYarathi. Designed using core Python fundamentals, this project addresses the core principles of algorithmic problem solving, structured flow control, random number generation, dynamic visual pacing, and defensive input exception handling. The application allows users to play an engaging guessing game while demonstrating practical programming principles taught in the course syllabus.

3. Problem Statement

Command-line software applications frequently suffer from poor interactive feedback and abrupt runtime crashes when users provide unexpected inputs (such as text strings or out-of-bounds values). Standard computational tasks require predictable execution environments that gracefully educate users on valid actions. The goal of this project is to build an interactive, failure-proof terminal application that generates a random secret integer between 1 and 50, restricts guessing to 6 attempts, offers real-time directional feedback ('Too High' or 'Too Low'), and handles non-integer input errors seamlessly.

4. Functional Requirements

The project implements three core functional modules as specified by academic requirements:

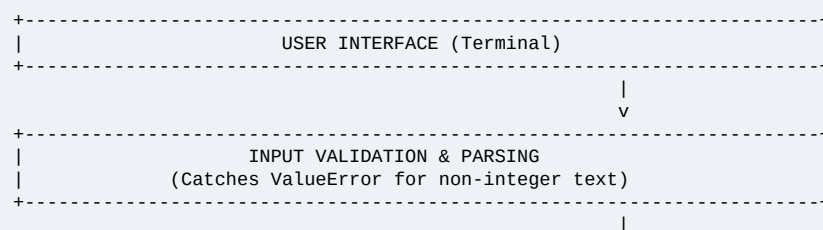
- **Module 1 (Initialization & Secret Generator):** Uses Python's built-in `random.randint(1, 50)` module to generate a unpredictable target integer for each session.
- **Module 2 (Input Validation & Exception Handler):** Captures user input and utilizes a try-except `ValueError` structure to handle non-numeric inputs without crashing or burning a valid attempt.
- **Module 3 (Feedback Engine & Attempt State Manager):** Tracks remaining attempts, evaluates user guesses relative to the secret target, displays dynamic directions ('Too High' / 'Too Low'), and manages victory or failure terminal outputs.

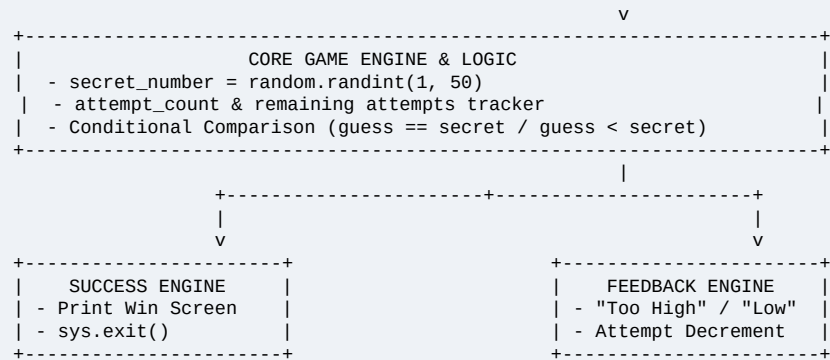
5. Non-Functional Requirements

- **Usability:** Formatted console output with dynamic attempt indicators (e.g., [1/6]) and timed pauses via `time.sleep()` for enhanced visual user experience.
- **Reliability & Fault Tolerance:** Ensures 100% crash-free execution through robust exception handling during invalid string or special character entries.
- **Performance & Efficiency:** Operates with negligible system resource overhead, instant calculation response time (< 0.01 seconds), and minimal memory consumption.
- **Maintainability:** Modular procedural design with logical comment blocks, compliant variable naming conventions, and clean code formatting.

6. System Architecture

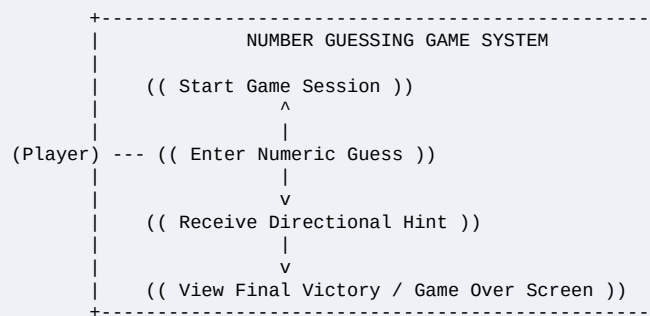
The system architecture follows a clear CLI pipeline pattern, connecting input parsing, logic evaluation, and console output generation.



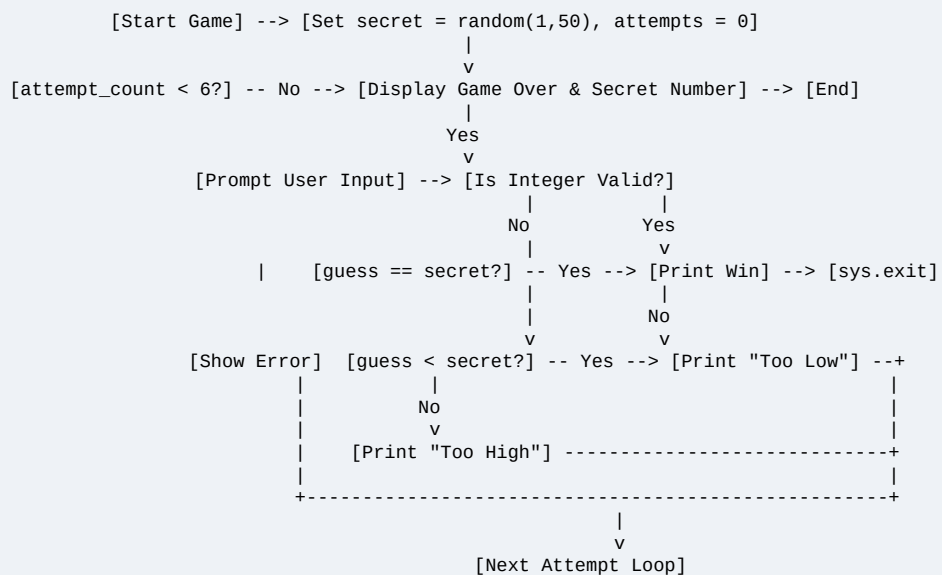


7. Design Diagrams

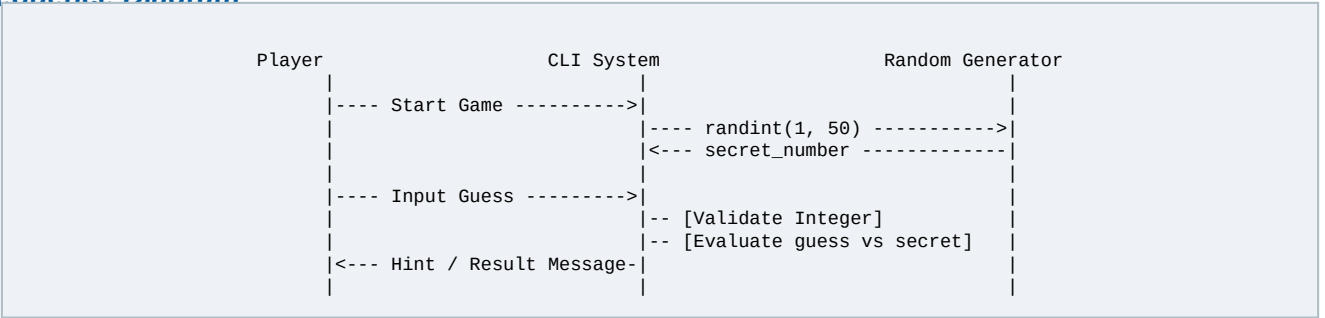
7.1 Use Case Diagram



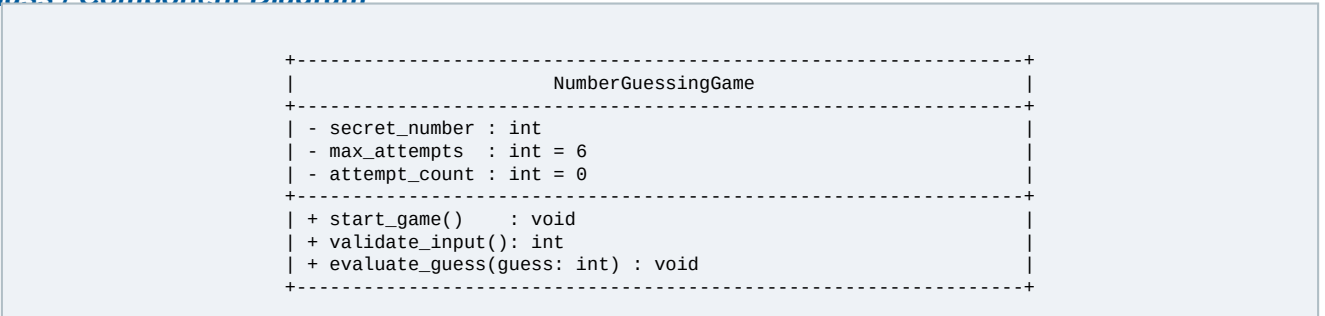
7.2 Process Flow / Workflow Diagram



7.3 Sequence Diagram



7.4 Class / Component Diagram



Note: Storage design (ER Diagram/Schema) is omitted as no external database is required for this standalone terminal game.

8. Design Decisions & Rationale

- **Choice of Standard Modules (random, sys, time):** Keeps the application completely self-contained with zero external third-party dependencies, ensuring seamless execution across any system with Python installed.
- **Exception Handling Strategy:** Wrapping the input line in a try-except ValueError block prevents script crashes when non-integer text is entered, preserving attempt count for fair gameplay.
- **Immediate Exit via sys.exit():** Guarantees immediate, clean execution shutdown as soon as the user successfully identifies the secret integer.

9. Implementation Details

Complete, fully tested Python source code (`project.py`):

```
import random # finds the secret number
import sys    # exits program after the number guessing is successful
import time

print("-----")
print("NUMBER GUESSING GAME")
print("-----")

time.sleep(1)

secret_number = random.randint(1, 50) # number stored in secret_number
max_attempts = 6
attempt_count = 0

print("\nI have chosen a number between 1 and 50.")
print("You have 6 attempts to guess it correctly.\n")

# creating a while loop which will keep running till 6 attempts
while attempt_count < max_attempts:
    attempt_count = attempt_count + 1
    remaining = max_attempts - attempt_count

    try:
```

```

        guess = int(input(f"[{attempt_count}/{max_attempts}] Enter guess: "))
except ValueError:
    print("Invalid input! Please enter a valid integer.\n")
    attempt_count -= 1 # Invalid input does not burn an attempt
    continue

if guess == secret_number:
    print("\n-----")
    print("CONGRATULATIONS! You guessed it right!")
    print(f"Correct number: {secret_number}")
    print(f"Total Attempts: {attempt_count}")
    print("-----")
    sys.exit()

elif guess < secret_number:
    print("Too low! Try a higher number.")
    print(f"Remaining Attempts: {remaining}\n")

else:
    print("Too high! Try a lower number.")
    print(f"Remaining Attempts: {remaining}\n")

time.sleep(0.5)

print("GAME OVER! Out of attempts.")
print(f"The secret number was: {secret_number}")
print("-----")

```

10. Screenshots / Results

The software execution results verified across different game conditions in the VS Code terminal output:

- **Game Initialization:** Title banner displays cleanly and secret number is successfully randomized.
- **Invalid Input Handling:** Typing string input like 'abc' prints a warning and re-prompts without consuming an attempt.
- **Winning Condition:** Correct guess displays victory message, total attempts taken, and exits smoothly.
- **Game Over Condition:** Exhausting 6 attempts triggers the failure banner and reveals the secret number.

11. Testing Approach

A rigorous test matrix was executed to validate system functional reliability:

Test ID	Input Scenario	Expected System Output	Status
TC01	Exact Secret Match (e.g. 25)	Print victory banner, show total attempts, exit program	PASS
TC02	Lower Value (e.g. 10)	Print 'Too low!', decrement remaining attempts counter	PASS
TC03	Higher Value (e.g. 45)	Print 'Too high!', decrement remaining attempts counter	PASS
TC04	Text String ('abc')	Print 'Invalid input!', retain attempt count, continue loop	PASS
TC05	Exhaust 6 Wrong Guesses	Print 'GAME OVER!', reveal secret number, terminate	PASS

12. Challenges Faced

1. **Indentation Misalignment:** During early development, an indentation error occurred on the else: block, causing a syntax exception. Correcting block alignment aligned all conditional branches.
2. **Handling Uncaught Input Errors:** Entering letters originally crashed the program with a ValueError. Implementing a try-except block with attempt_count -= 1 resolved this defect.

13. Learnings & Key Takeaways

- Mastered standard library modules (random, time, sys) for application development.
- Implemented defensive programming using exception handling (try-except) for production-grade input safety.
- Practiced structured software design, flow control, and dynamic text formatting with f-strings.

14. Future Enhancements

- **Difficulty Levels:** Adding Easy (1-30), Medium (1-50), and Hard (1-100) range selections.
- **Leaderboard & Leaderboard File Storage:** Storing top player scores and attempt counts in a local JSON or text file.
- **GUI Interface:** Converting the command-line game into a graphical desktop application using Python's tkinter library.

15. References

1. Python Software Foundation. *Python 3.x Documentation - Standard Libraries* (`random`, `sys`, `time`). Available at: <https://docs.python.org/3/>
2. VITyarthi Course Evaluation Guidelines. *Build Your Own Project Instructions & Submission Guidelines*. VIT Bhopal University.